

Yuyang Cheng

jrm9ga@virginia.edu | 434-235-0892 | [Google Scholar](#) | [LinkedIn](#)

EDUCATION

University of Virginia, School of Data Science Sep 2025 – Present
Ph.D. student in Data Science | Advisor: Chirag Agarwal Charlottesville, Virginia

Research Interests: Trustworthy AI Agents, AI Safety & Interpretability, Multi-modal Learning

Peking University, School of Software & Microelectronics Sep 2022 – June 2025
Master of Engineering in Computer Technology Beijing, China

University of Electronic Science and Technology of China (UESTC) Sep 2018 – June 2022
B.A. in Translation & B.Eng. in Computer Science Chengdu, China

GPA: 3.88/4.00; Computer Science GPA: 4.0/4.0

Rank: Top 1 in Computer Science and Top 5 in Translation

PUBLICATIONS

* Equal contribution.

- [1] **MEA: A Reward-Driven Multi-Agent System for Faithful Model Explanations** [\[Paper\]](#)
Yuyang Cheng, Raghav Kaushik Ravi, Srivarshinee Sridhar, Sriparna Saha, Akash Ghosh, Chirag Agarwal
Submitted to ICLR 2027, The International Conference on Learning Representations
- [2] **Aligning Safety-Critical Steps in Mobile Agents via On-Policy Self-Distillation**
Chengjie Liu*, **Yuyang Cheng***, Chenhui Cui, Tao Li, Wenjun Wu, Weiping Wen, Rubing Huang, Chirag Agarwal
In submission to FSE 2027, The ACM International Conference on the Foundations of Software Engineering
- [3] **Observing Micromotives and Macrobehavior of Large Language Models** [\[Paper\]](#)
Yuyang Cheng, Xingwei Qu, Tomas Goldsack, Chenghua Lin, Chung Chi Chen
*Accepted by AACL 2025, **Oral**, Proceedings of the 14th International Joint Conference on Natural Language Processing*
- [4] **MAP-Neo: Highly Capable and Transparent Bilingual Large Language Model Series** [\[Paper\]](#)
Ge Zhang*, Scott Qu*, Jiaheng Liu*, Chenchen Zhang*, Chenghua Lin*, Chou Leuang Yu, Danny Pan, **Yuyang Cheng**, et al.
- [5] **Chinese Tiny LLM: Pretraining a Chinese-Centric Large Language Model** [\[Paper\]](#)
Xinrun Du*, Zhouliang Yu*, Songyang Gao*, Ding Pan, **Yuyang Cheng**, et al.
Accepted by COLM 2024, Conference On Language Modeling
- [6] **CMMMU: A Chinese Massive Multi-discipline Multimodal Understanding Benchmark** [\[Paper\]](#)
Ge Zhang*, Xinrun Du*, Bei Chen*, Yiming Liang, Tongxu Luo, Tianyu Zheng, Kang Zhu, **Yuyang Cheng**, et al.

RESEARCH EXPERIENCE

MEA: Multi-Agent System for Faithful Model Explanations Nov 2026 – Sep 2026
Chirag Agarwal, University of Virginia Advisor: Chirag Agarwal

- Developed a multi-modal multi-agent framework that makes post-hoc explanations accessible to non-experts. Trained with faithfulness rewards and a modality-adaptive penalty, MEA outper-

formed post-hoc, agentic, and closed-source baselines on 12 datasets, improving faithfulness over the untrained backbone by 28% (tabular), 21% (text), and 34% (vision).

- Introduced question types spanning feature attribution, counterfactual reasoning, and spurious feature detection, each paired with a perturbation-based faithfulness metric.

Safety Alignment for Mobile Agents via On-Policy

June 2026 – Sep 2026

Self-Distillation

University of Virginia

Advisor: Chirag Agarwal

- Proposed a lightweight framework that builds state-transition probability graphs from agent trajectories, scores node risk via reverse risk propagation, and applies on-policy self-distillation only at critical steps identified by risk divergence between candidate branches.
- Substantially reduced attack success rates across multiple mobile agent models while preserving task utility and training significantly faster than baseline alignment methods.

Observing Micromotives and Macrobehavior of Large Language Models

July 2024 – Oct 2024

University of Manchester

Advisor: Chenghua Lin

- Developed a computational social science approach using agent-based modeling to study societal segregation dynamics, incorporating NLP techniques to analyze how algorithmic decision-making may influence group interactions and macro-level social patterns.
- Found that algorithmic suggestions in social simulations can amplify existing societal segregation trends, highlighting the importance of interdisciplinary research in computational social science and responsible AI development.

HK Generative AI R&D

Nov 2023 – Mar 2024

The Hong Kong University of Science and Technology

Advisor: Jie Fu

- *Chinese Tiny LLM (COLM)*: Led data processing and group work coordination as group leader; adapted the filtering rules for Chinese datasets, since previous rules mainly targeted English data, and confirmed appropriate thresholds through analysis of sampled documents.
- *MAP-Neo*: Core contributor to pre-training data collection and the filtering pipeline; participated in Optical Character Recognition and relative calibration; proposed strict data filtering rules to optimize linguistic quality and reduce biases, retaining only 4% and 19% of the original corpus.
- *CMMMU*: Contributed to dataset construction and annotation quality control; standardized data formats and annotation guidelines.

National Key R&D Program

Apr 2021 – July 2021

University of Electronic Science and Technology of China

Advisor: Mengshu Hou

- Participated in project “Distributed Data Space Construction and System for Manufacturing Value Chain”, studied opinion extraction, topic clustering, and sentiment analysis from collected data.

PROFESSIONAL EXPERIENCE

China Mobile Research Institute, Artificial Intelligence and Smart Operations Center

Apr 2025 – Aug 2025

Research Scientist Intern

Beijing

- Led the personalization project of AI Chatbot, designed a seed-free pipeline to synthesize Chinese user personas for personalized LLM dialogue data, combining sampled attributes with LLM-based consistency filtering.

INF Technology (Shanghai) Co., Ltd.

Apr 2024 – Dec 2024

Algorithm Intern

Shanghai

- Led the Text2SQL task training for the pre-trained model, responsible for dataset validation, algorithm framework design and optimization, model training, and evaluation.

- Contributed to the design and implementation of the Reinforcement Learning training phase for the pre-trained model.

Common Language – Gender Equality NGO

Mar 2024 – July 2024

Field Development Intern

Beijing

- Spearheaded the AI Empowerment Workshop for Grassroots Nonprofit Organizations.
- Applied machine learning algorithms to analyze demographic data on Chinese citizens' attitudes toward LGBTQ+ issues.

Beijing Milian Science and Technology Co.

May 2023 – July 2023

Assistant Research Engineer

Beijing

- Participated in the construction and tuning of a chatbot model, responsible for the preliminary construction of a multi-dimensional evaluation model for Natural Language Generation.

AWARDS & HONORS

- Outstanding Graduate, UESTC
- Outstanding Students Delegation Team, UESTC
- Outstanding Student Scholarship (2×), UESTC
- Honorable Mention, 2021 Mathematical Contest in Modeling
- Provincial 2nd Place, 11th National E-Commerce Innovation, Creativity & Entrepreneurship Contest

TEACHING & SERVICES

Teaching Assistant, University of Virginia: How to Train Your LLM
 Advanced ML II: Methods & App
 ML I: Intro to Predictive Modeling
 Data Eng II: Big Data Systems